

PANKHURI KULSHRESTHA

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Profile

Dedicated and highly skilled Computer Vision Engineer with 8 years of experience and a strong background in computer science, mathematics, and statistics. Proficient in developing AI-driven applications ensuring real-time performance on limited computational resources. Adept at utilizing various machine learning frameworks and libraries to design and deploy scalable and efficient solutions. Adept at architecting scalable, secure, and cost-effective cloud solutions in AWS, Azure, and Google Cloud.

Key Skills

- Python, C++
- SQL
- Machine Learning Frameworks: PyTorch, TensorFlow, ONNX, Scikit-learn
- Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
- Docker, Kubernetes, Jenkins
- Git, GitHub
- Mathematics and Statistics: Linear Algebra, Probability Theory, Statistical Analysis
- Deep Learning: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers

Achievements and Certifications

- [Earned Google Developer Badge in Machine Learning Bootcamp, Oct 2022](#)
- [Tensorflow Certification, Oct 2022](#)
- [Ranked within 14% Hub map Organ Segmentation Kaggle competition, 22 Sept 2022](#)
- [Coursera Deep Learning Specialization, 15 Aug 2022](#)
- [Post Graduate Program in Artificial Intelligence and Machine Learning, 31 Dec 2018 – 31 Dec 2019](#)

Open-Source Contributions

- [Mindtrace](#) – Open-Source AI Infrastructure Framework:
Contributing to the development of modular micro-services for machine learning orchestration, observability, and agent-based automation.

Technical Skillset

- Proficient in researching, developing, and deploying state-of-the-art computer vision solutions.
- Experience in training SOTA 2D models/LVM for object classification, detection and segmentation like Dino-v2, Yolo10, Deeplab
- Experienced in development of compact, low-power, and cost-effective solutions for deploying computer vision capabilities in IoT applications.
- Experience in training 3D Machine Learning models for Point Cloud Data, with a focus on 3D semantic scene segmentation like SPVNAS/PointNet, and generative 3D scene modelling for Asset Management scenarios.
- Design AI algorithms for various applications including OCR, object detection, and object segmentation.
- Experience in training ML models with frameworks like Tensorflow, PyTorch
- Facilitate design and deployment of necessary vision hardware equipment for capturing image data.
- Monitor and maintain deployed models, ensuring accuracy and scalability.
- Experienced in handling large datasets and implementing data preprocessing techniques to extract meaningful insights.
- Experience in Data-centric AI augmentation or auto labelling.
- Ability to be innovative and propose novel ideas.
- Strong communication, documentation, presentation, and interpersonal skills

Work experience

Mindtrace, Manchester, Machine Learning Research Engineer

Jan 2023 – Present

- Worked with clients for asset inspection to automate their process of surveying and assessing the physical condition of assets by making their applications AI-assisted. Utilized advanced computer vision techniques to replace repetitive monitoring tasks, significantly improving efficiency and accuracy, ultimately enabling clients to transition from manual inspections to automated systems, reducing operational costs and increasing the precision of asset condition assessments.
- Utilized the latest self-supervised learning model, DINO v2 features, to train on large-scale unlabelled datasets, resulting in a significant reduction in the need for manual annotation and improving model performance.
- Fine-tuned pre-trained DINO v2 models on specific tasks using Lora, achieving a 95% accuracy on our target dataset.
- Developed and trained advanced 3D deep learning models for image segmentation, specifically focusing on Point Cloud data. Worked on enhancing feature pre-processing techniques, improving loss and metric functions, and training generalized point cloud segmentation models.
- Collaborated with cross-functional teams to integrate these models into our product pipeline, improving the overall user experience and driving an increase in customer satisfaction.
- Presented findings and progress to stakeholders, providing clear explanations of complex technical concepts and their business implications.
- Demonstrated expertise in deploying models in cloud environments like Azure, AWS and GCP for model inference, resulting in improved efficiency and scalability of our data processing pipeline.

Edcast, India, Computer Vision Engineer

June 2021 – Dec 2021

- Implemented Text detection and extraction using Google Vision API and Tesseract OCR in images
- Utilized FLANN based Matcher and SIFT algorithms for object identification.
- Worked on AWS Services like Lambda/API Gateway/S3 with serverless framework.

IQVIA, India, Software Engineer. September 2019 – May 2021

- Worked on a computer-implemented method for classifying and interpreting life science documents for a Content Management System. The solution designed for classifying documents involved text analysis of digitized representations to identify raw words in the text, construct analysis to identify document elements and relative spatial positioning of document elements on a page, image analysis to identify images and process the identified images to extract text.
- Worked on Document Layout Analysis models comparing Faster RCNN and Mask RCNN
- Developed pipelines for text Data Cleansing, feature extraction, model development and deployment using Gitlab, Kubernetes and Docker.
- Created Kubernetes job to deploy Machine Learning models inside Kubernetes cluster and mount Persistent Volumes using Helm.
- Developed Microservices for model predictions and inference.
- Quantified and tracked the classification algorithms using F1 score, Precision and Recall

Bank of Paribas, India, Software Engineer

Sept 2017–Sept 2019

- Worked on fraud detection projects, utilizing machine learning algorithms and statistical techniques to identify and prevent fraudulent activities.
- Developed and implemented anomaly detection algorithms to detect unusual patterns in transaction data
- Worked on enabling deployment process for Bank Applications to use Continuous Integration and Continuous Delivery Platform using Jenkins and TeamCity.

Education

University Of Essex, United Kingdom, MSc (Hons) Artificial Intelligence

Jan 2022 – Jan 2023

Rajiv Gandhi Proudhyogiki Vishwavidyalaya, India, Bachelor of Engineering in Electronics and Comm.

May 2007 - May 2011

Publications

Privacy-Preserving Self-Learning Chatbot with Federated Intelligence, IEMTRONICS 2025

Evaluating Dialogue Adaptability: a Comparative Study of Self-Feeding Mechanisms in Federated and Centralized Chatbot Architectures, SAIA-2025

References

Dr Sefki Kolozali, Lecturer, University of Essex (CSEE Department), email: sefki.kolozali@essex.ac.uk

Vikrant Sharma, Tech Lead, Mindtrace, email: vikrant@mindtrace.ai